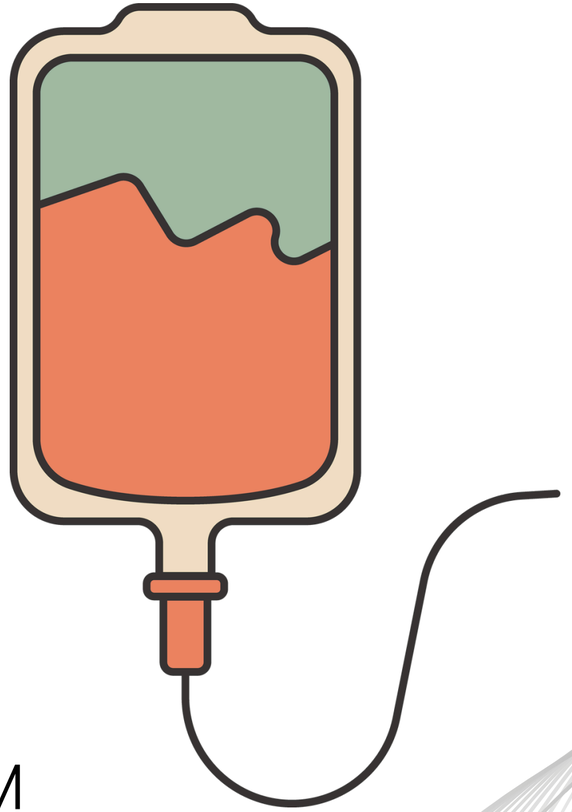


IIIV

PRESENTED BY TEAM PHLEGM

Multisensor failure-mode detection for gravity-based infusion setups

USER STORY

- Changing flow rate affects delivery of medication
- Understaffed clinics and hospitals in low-income areas
- Too many patients to handle individually (overworked)
- Lack of categorization of the problem with the transfusion
- **36% of IV catheter transfusions result in failure** before therapy completion.
- **72% to 99% of clinical alarms are false alarms** leading to alarm fatigue.

PROBLEM STATEMENT

There is a monitoring gap in gravity-based IV systems.

No automated differentiation between:

- Empty bag
- Occlusion
- Leak/disconnection

Low income and/or understaffed hospitals may not be able to resolve issues with IV bags in a timely manner before severe damage is done to patients.

WHY EXISTING DROP MONITORS AREN’T ENOUGH.

Background

- IV pump is gravity-based
- Sensitive to patient movement
- Must understand flow rate for proper medication transfusion

Monidrop Device

- Detect drop rate only
- Threshold-based alarms
- No failure classification
- No sensor redundancy

OUR SOLUTION: IIIV

A clip-on, multisensor gravity IV integrity monitor.

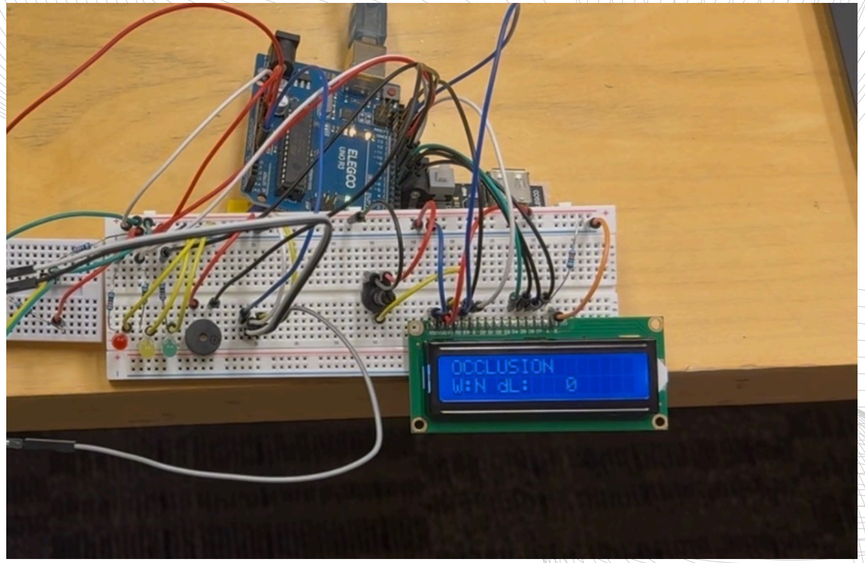
Two sensing channels:

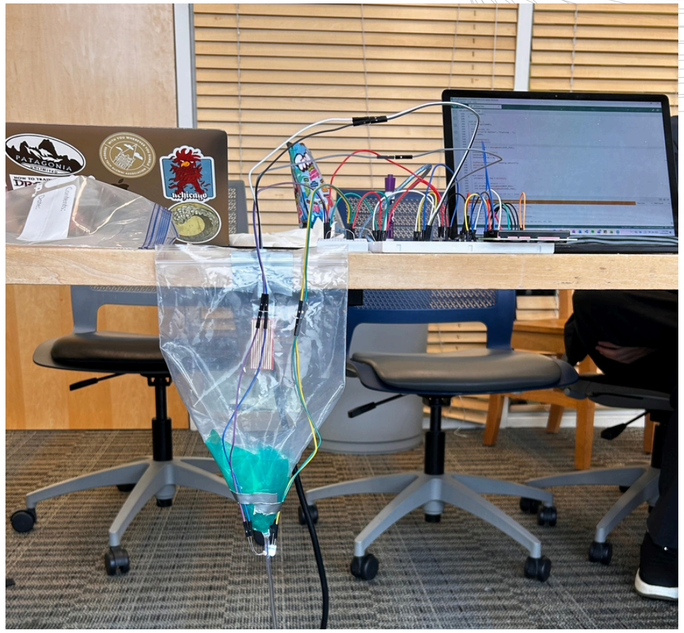
1. Reservoir depletion trend (water level sensor)
2. Flow activity signal (photoresistor)

Fusion logic classifies:

1. Normal
2. Leak/External Influence
3. Occlusion

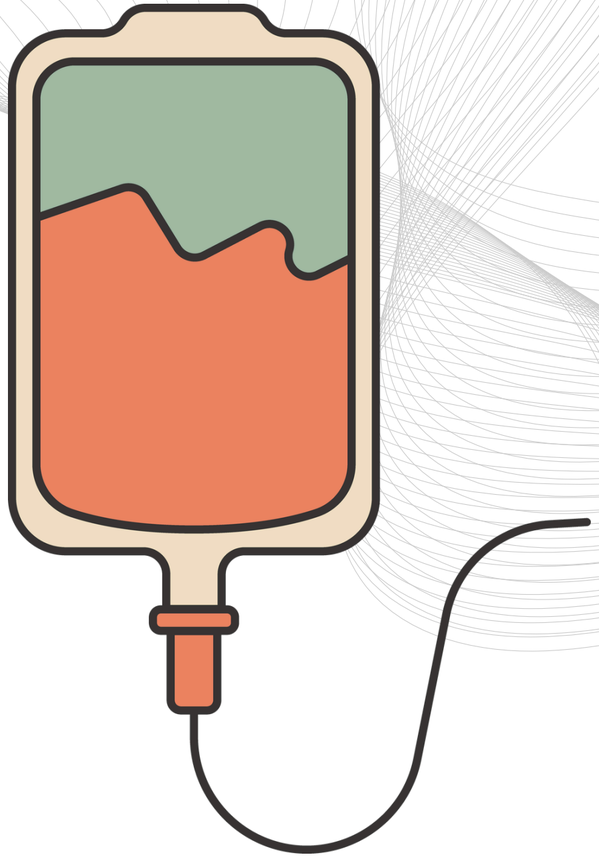
→ Connected to a singular, easy-to-use interface.



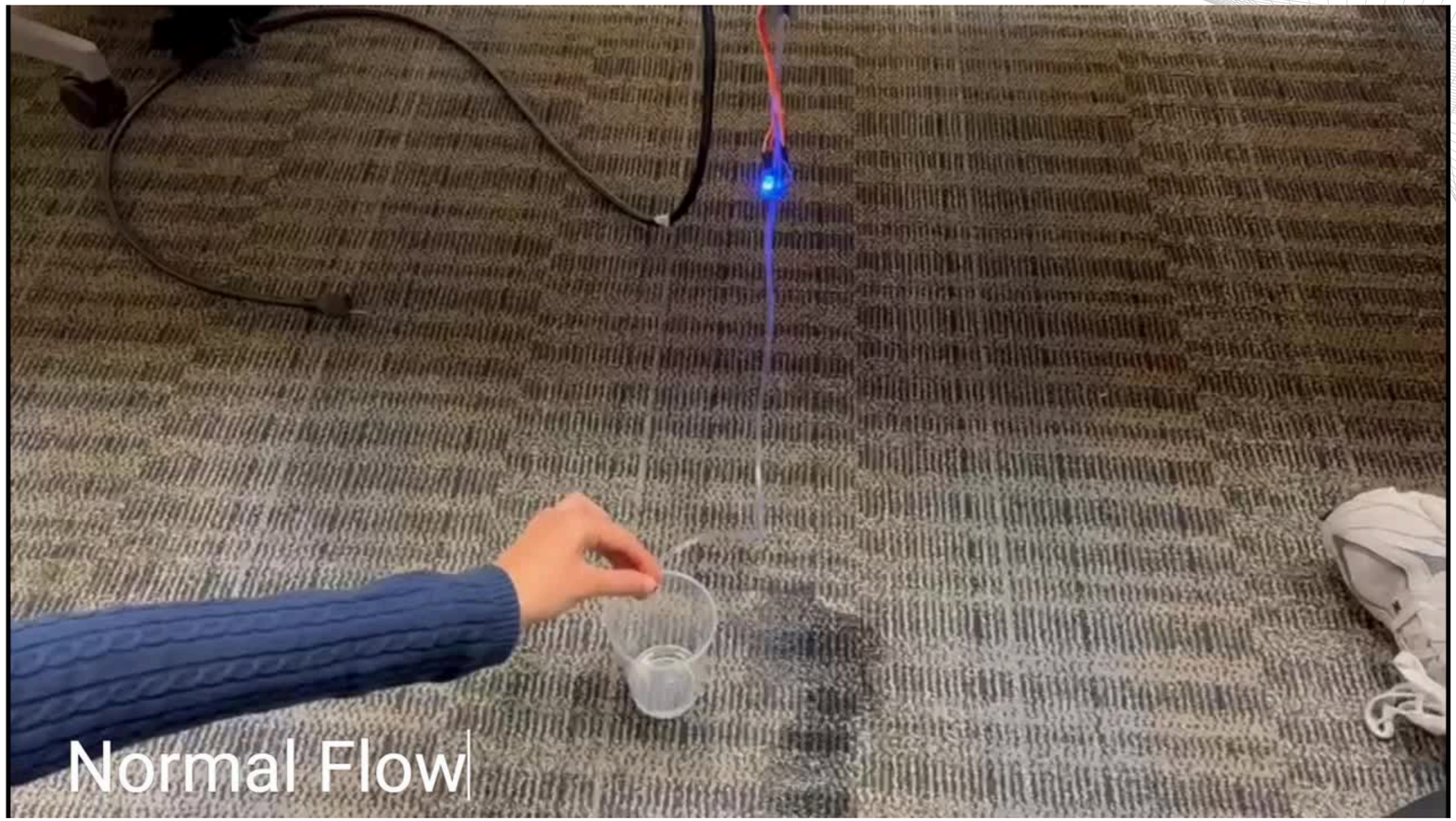


SENSOR FUSION

Condition	Water Level Sensor	Photoresistor
Normal	Decreasing	Active
Occlusion	Stable	Inactive
Leak/External	Decreasing	Inactive



DEMOS



CLINICAL IMPACTS

1. Adds intelligence to gravity IV systems
2. Low-cost components
3. No tubing modifications required
4. Reduces nurse cognitive load
5. Supports low-resource and ward-based settings

COST BREAKDOWN

Prototype Cost

- Water Level Sensor: $1
- Photo Resistor: <$1
- LED: <$2
- Wires <$0.50
- Resistors: <$1
- Potentiometer <$1
- Buzzer <$1
- LED <$0.50
- Microcontroller $10
- Breadboard <$2

Total: approx. $20

Competitor Costs

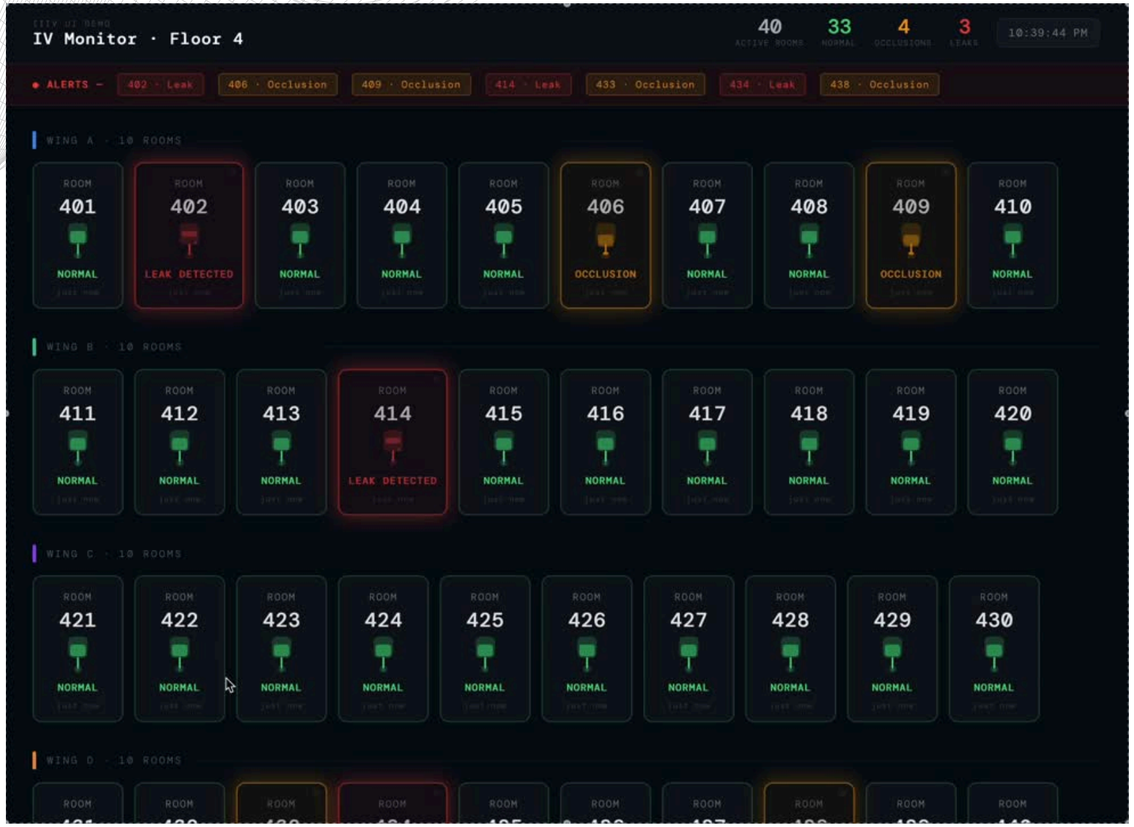
- DripAssist: approx. $400 (Couperus, 2019)
- Monidrop: approx. >$500

✳ Note this is a prototype vs finished product cost comparison

FEASIBILITY & SCALABILITY

- **Attachable sensor system** allows for function on every kind of gravity-based IV bag.
  - Much more affordable than IV systems with tech built into it.
- **Universal code and calibration** features ensures functionality
  - Can be ran offline and directly loaded onto microcontroller
- **Clean, simple software platform** enables nurses to monitor floors at a glance
  - Multiple systems can be controlled through **one** central device

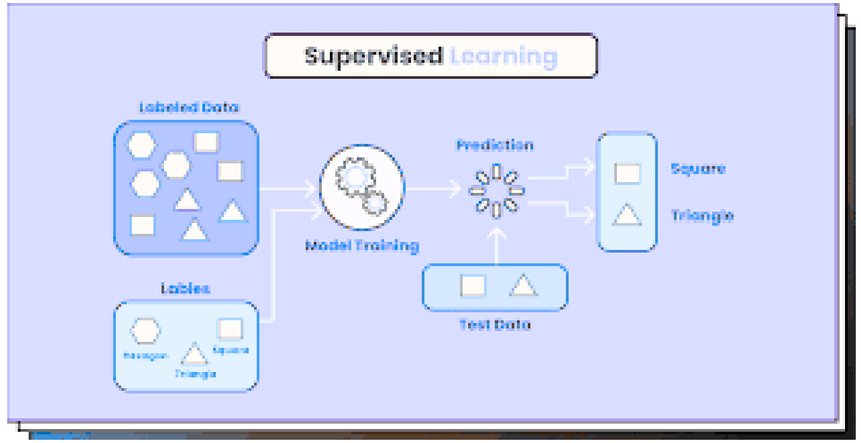
USER INTERFACE



- Connects all IV + sensor systems into **one clean, easy-to-use interface**
- Occlusions + Leaks detected will immediately ping the room it's in
- Allow medical personnel to **immediately respond** to patients in need + resolve issues after

FUTURE STEPS

- Movement ML Model
- Predictive Model
- Improve photoresistor reading accuracy
- Quality of water level sensor
- Length of water level sensor
- Assembly + Casing



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